

Representation Dependence of Optimal Cut Placement in Quasiprobability Circuit Cutting

Phung Trong Hung* and Huong Bui†

Faculty of Information Technology, Ho Chi Minh City University of Technology (HUTECH), Ho Chi Minh City, Vietnam

(Dated: October 7, 2026)

Circuit cutting incurs a sampling overhead when two-qubit interactions cross fragment boundaries. Different gate-level representations can implement the same logical unitary while assigning different quasiprobability-decomposition (QPD) costs to those interactions. We ask whether this representation choice alone can change which feasible partition minimizes the modeled cutting overhead. A complication is degeneracy: when several partitions are optimal, comparing one solver-returned partition from each representation can falsely suggest that the optimum changed. We therefore compare the optimal sets themselves by evaluating, in a target representation, the best partition among those that are optimal in the source representation. The resulting cross-representation regret is zero exactly when the two optimal sets overlap. We derive simple perturbation and stability bounds for this quantity. On quantum phase-estimation benchmarks, an $n = 14$, $K = 2$ pair gives a genuine modeled-overhead ratio $R = 729$; exhaustive enumeration of all 1,716 balanced bipartitions confirms unique, disjoint optima, and MQT QCEC independently verifies the circuit equivalence up to global phase. By contrast, six apparent quantum-Fourier-transform changes from single solver outputs disappear under the set-level comparison, and exhaustive enumeration confirms a shared optimum in every case. For a fixed representation, the independent gate-level QPD objective is exactly a capacitated weighted graph-partitioning objective in the log domain. Thus a semantics-preserving representation change can alter the resource-optimal cut placement even when the logical unitary and feasible fragment space are fixed.

I. INTRODUCTION

Near-term quantum processors remain limited in qubit count, fidelity, and connectivity [1]. Circuit cutting addresses the width constraint by replacing selected non-local operations with combinations of fragment-local operations and reconstructing the target quantity by classical post-processing [2–4]. In quasiprobability-based gate cutting, each severed two-qubit interaction contributes a gate-dependent sampling factor. The total overhead is multiplicative, so two placements with the same number of cuts can have very different costs.

The same logical circuit can also have several gate-level representations. Basis translation or decomposition can preserve the unitary up to global phase while changing the two-qubit gates that appear on each pair of logical wires. Under a gate-dependent QPD model, the corresponding interaction weights therefore change. The question studied here is whether that change can move the resource-minimizing cut placement when the logical wires, fragment count, and capacity constraints are held fixed.

There is a subtle degeneracy issue. If representation a has several optimal placements, a solver may return one that is suboptimal under representation b even though another a -optimal placement is also optimal under b . Comparing one returned solution from each representation can therefore create a false representation effect. We avoid this by optimizing over the source-optimal region

rather than using one arbitrary returned placement. Section III defines the resulting cross-representation regret, derives an ℓ_1 perturbation bound and a sufficient stability condition, and identifies positive rescaling of all interaction weights as an invariant case.

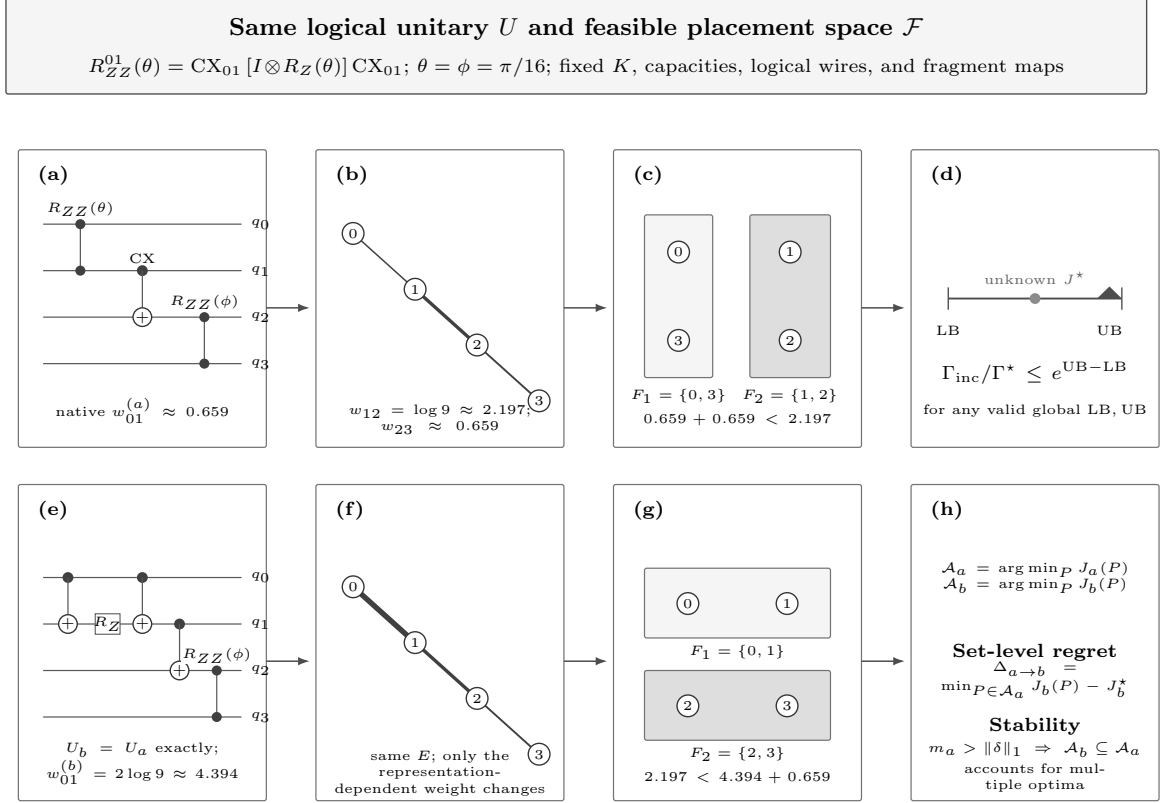
The numerical study records a semantic check before comparing representations, with the check method stored alongside each result. Exact-QPE benchmarks contain strict placement changes, including an $n = 14$, $K = 2$ instance for which transporting a native-QPD-optimal placement to the CX-normalized representation increases the modeled overhead by a factor of 729. An independent exhaustive audit of all 1,716 balanced bipartitions finds one optimum under each representation and zero overlap between the two argmin sets. The second $n = 14$ QPE change, at $K = 3$, is likewise confirmed by exhaustive enumeration of 126,126 near-balanced partitions; together with the exact $n = 6$, $K = 2$ record, all three QPE changes therefore have exhaustive support. In contrast, six large QFT regrets obtained by comparing one solver-selected optimum from each representation vanish under the set-level cross-evaluation; exhaustive enumeration of all six corresponding feasible spaces independently confirms a shared optimum in every case. These two outcomes provide positive and negative tests of the same set-level criterion.

For a fixed representation, the independent gate-level QPD objective is a capacitated weighted graph-partitioning problem in the log domain. This reduction supplies the optimization model used in the experiments and gives a direct interpretation of valid global lower and upper bounds: if $LB \leq OPT \leq UB$, then the incumbent overhead is at most $\exp(UB - LB)$ times the modeled

* Contact author: phungtronghung0808@gmail.com

† bd.huong@hutech.edu.vn

Semantics-preserving representation change and cut placement



Four-qubit witness with $\theta = \phi = \pi/16$: identity exact; edge width denotes independent-QPD log cost.

FIG. 1. A four-qubit example of the mechanism studied in this work. Panels (a)–(d) show representation a , its weighted interaction graph, an optimal balanced bipartition, and the translation of valid log-objective bounds into an overhead ratio. The displayed instance fixes $\theta = \phi = \pi/16$, for which Qiskit Addon Cutting gives $\log \rho_{RZZ(\pi/16)} \approx 0.659$ and $\log \rho_{CX} = \log 9 \approx 2.197$. Panels (e)–(h) replace $R_{ZZ}^{01}(\theta)$ by the exact identity $CX_{01}[I \otimes R_Z(\theta)]CX_{01}$. The logical unitary, logical wires, and feasible partition space are unchanged, while the independent-QPD weight on edge (0,1) increases from approximately 0.659 to $2 \log 9 \approx 4.394$. Consequently the three balanced bipartitions can be checked directly: the optimum changes from $\{0, 3\}|\{1, 2\}$ to $\{0, 1\}|\{2, 3\}$. Panel (h) shows the set-level comparison used to avoid arbitrary optimizer tie breaking. ChatGPT (OpenAI, GPT-5.6 Sol) assisted with schematic layout revision; the authors specified and verified the circuit identity, graph weights, partitions, labels, and final LaTeX figure.

optimum. The graph-partitioning and mixed-integer machinery itself is standard [5–7].

Our question differs from work that searches a richer cut space for a fixed circuit. Brandhofer *et al.* [8] optimize mixed gate and wire cuts, Nakamura *et al.* [9] study scalable multi-fragment partitioning and sampling bounds, and Fröhler *et al.* [10] consider gate, wire, and joint gate-group cuts. Here the fragment space is fixed and the circuit representation changes. The primary cost model is the independent gate-level QPD model implemented by Qiskit Addon Cutting 0.10.0 [11].

II. CIRCUIT CUTTING AND REPRESENTATION-DEPENDENT QUASIPROBABILITY COSTS

A. Independent-QPD circuit cutting

Gate cutting replaces selected two-qubit channels by quasiprobability decompositions (QPDs) over fragment-local channels [2, 12]. For a cuttable channel $\mathcal{U}_g = \sum_i a_{g,i} \mathcal{E}_{g,i}$, the corresponding independent sampling overhead is

$$\rho_g = \left(\sum_i |a_{g,i}| \right)^2. \quad (1)$$

For a partition P with crossed-gate set $S(P)$, independent sampling across gates gives

$$\Gamma(P) = \prod_{g \in S(P)} \rho_g, \quad J(P) = \log \Gamma(P) = \sum_{g \in S(P)} \log \rho_g. \quad (2)$$

This objective matches the independent gate-level model in Qiskit Addon Cutting 0.10.0 [11]; it is not a model of joint decompositions, finite-shot variance, or hardware noise. Because the QPD coefficient one-norm is at least one, $\rho_g \geq 1$ and all log weights are nonnegative. We validated the gate costs against the Qiskit QPD registry: $\rho_{CX} = \rho_{CZ} = 9$, $\rho_{\text{ISWAP}} = \rho_{\text{DCX}} = 49$, $\rho_{CS} = 3 + 2\sqrt{2} \approx 5.828$, and $\rho_{RZZ(\theta)} = (1 + 2|\sin \theta|)^2$ for numerically bound rotations.

B. Log-domain interaction graph

Let $V = \{0, \dots, n-1\}$ be the circuit qubits, and let G_{ij} contain the two-qubit gate occurrences acting on pair (i, j) . We assign the nonnegative interaction weight

$$w_{ij} = \sum_{g \in G_{ij}} \log \rho_g. \quad (3)$$

For any fragment map $f : V \rightarrow \{1, \dots, K\}$, the objective becomes

$$J(P) = \sum_{i < j} w_{ij} \cdot \mathbb{I}(f(i) \neq f(j)). \quad (4)$$

The binary bisection identity $\mathbb{I}(f(i) \neq f(j)) = |z_i - z_j|$ is recovered when $K = 2$.

Proposition 1 (Objective equivalence). *Under independent gate-level QPD, minimizing (2) is equivalent to minimizing (4).*

Proof. Every $g \in G_{ij}$ is cut exactly when $f(i) \neq f(j)$. Regrouping the gate-level sum by qubit pair yields (4).

C. Representation changes and the common edge universe

For the semantics-preserving transformations considered here, basis translation and native-gate decomposition produce circuits implementing the same unitary up to global phase while altering the two-qubit interaction multigraph, and hence the weights (3). Let a and b be two semantically equivalent representations of the same logical circuit on the same logical wire set V , with interaction edge sets E_a and E_b . We define the common edge universe $E = E_a \cup E_b$ and zero-pad the log-weights, $w_e^{(a)} = 0$ for $e \notin E_a$ and $w_e^{(b)} = 0$ for $e \notin E_b$, so that both representations are compared over a single feasible partition space. Because the paired representations share the same logical wire set and declared capacity constraints,

the same partition map is feasible for both within the supported independent-QPD model; only the induced interaction weights change. Section III makes precise the sense in which a change from a to b can move the optimal placement.

The four-qubit construction in Fig. 1 gives an analytic example. Set $\theta = \phi = \pi/16$, write $x = \log \rho_{RZZ(\pi/16)} \approx 0.658867$ and $c = \log 9 \approx 2.197225$, and impose a balanced $K = 2$ partition. There are only three unlabeled bipartitions. In representation a , $\{0, 3\}|\{1, 2\}$ cuts the two RZZ edges and has cost $2x \approx 1.317735 < c$, while $\{0, 1\}|\{2, 3\}$ cuts only the central CX edge and has cost c ; the third bipartition cuts all three edges. After replacing $R_{ZZ}^{01}(\theta)$ by the exactly equivalent $CX_{01}[I \otimes R_Z(\theta)]CX_{01}$, the $(0, 1)$ weight becomes $2c$, so the first partition costs $2c + x \approx 5.053317$ whereas the second remains at c . The unique optimum therefore changes although the logical unitary and feasible partition set are unchanged.

III. REPRESENTATION-SENSITIVE OPTIMAL CUT PLACEMENT

A. Cross-representation regret over optimal sets

Consider the fixed logical qubit set $V = \{0, \dots, n-1\}$ evaluated over a common feasible partition space $\mathcal{F}$ defined by a fixed fragment count K and capacity bounds. For any partition $P \in \mathcal{F}$, the log objectives of the two representations are $J_a(P) = (w^{(a)})^\top z(P)$ and $J_b(P) = (w^{(b)})^\top z(P)$, where $z(P) \in \{0, 1\}^{|E|}$ is the cut-indicator vector over the common edge universe E of Sec. II C. Let $\mathcal{A}_a = \arg \min_{P \in \mathcal{F}} J_a(P)$ and $J_b^* = \min_{P \in \mathcal{F}} J_b(P)$. We define the set-level cross-representation regret and corresponding sampling-overhead ratio by

$$\Delta_{a \rightarrow b} = \min_{P \in \mathcal{A}_a} J_b(P) - J_b^*, \quad R_{a \rightarrow b} = \exp(\Delta_{a \rightarrow b}). \quad (5)$$

The minimum over $\mathcal{A}_a$ reports positive regret only when every a -optimal placement is suboptimal under b . When $R_{a \rightarrow b} = 1$, at least one optimal partition under representation a remains optimal under b ($\mathcal{A}_a \cap \mathcal{A}_b \neq \emptyset$). When $R_{a \rightarrow b} > 1$, every optimal partition under a is strictly suboptimal under b . In general $R_{a \rightarrow b} \neq R_{b \rightarrow a}$.

B. Representation-regret bound

The following perturbation statements are exact model-level results; the medium-tier numerical evaluations of Sec. V are solver-tolerance quantities. Let $\delta = w^{(b)} - w^{(a)} \in \mathbb{R}^{|E|}$ be the interaction-weight perturbation vector over the edge universe E .

Proposition 2 (Representation-regret bound). *For any common feasible partition space $\mathcal{F}$, $0 \leq \Delta_{a \rightarrow b} \leq \|\delta\|_1$, and $R_{a \rightarrow b} \leq \exp(\|\delta\|_1)$.*

Proof. Let $P_a \in \arg \min_{P \in \mathcal{A}_a} J_b(P)$ and $P_b \in \mathcal{A}_b$. Then $J_b(P_a) - J_b(P_b) = [J_a(P_a) - J_a(P_b)] + \delta^\top [z(P_a) - z(P_b)]$. Because P_a is J_a -optimal, $J_a(P_a) - J_a(P_b) \leq 0$. Since $\|z(P_a) - z(P_b)\|_\infty \leq 1$, we obtain $\Delta_{a \rightarrow b} \leq |\delta^\top [z(P_a) - z(P_b)]| \leq \|\delta\|_1$.

C. Stability and scaling invariance

Let $m_a = \min_{P \in \mathcal{F} \setminus \mathcal{A}_a} [J_a(P) - J_a^*]$ denote the optimality margin of representation a , with the convention $m_a = +\infty$ if $\mathcal{F} \setminus \mathcal{A}_a = \emptyset$. Define m_b analogously for representation b .

Proposition 3 (Sufficient representation stability). *If $m_a > \|\delta\|_1$, then $\mathcal{A}_b \subseteq \mathcal{A}_a$, and consequently $R_{a \rightarrow b} = 1$.*

Proof. For any $P \in \mathcal{A}_a$ and $Q \in \mathcal{F} \setminus \mathcal{A}_a$, $J_a(Q) - J_a(P) \geq m_a$. The perturbation changes the objective difference by at most $|\delta^\top [z(Q) - z(P)]| \leq \|\delta\|_1$. If $m_a > \|\delta\|_1$, then $J_b(Q) - J_b(P) \geq m_a - \|\delta\|_1 > 0$, so no non-optimal partition $Q \notin \mathcal{A}_a$ can achieve a smaller or equal J_b objective than P . Thus $\mathcal{A}_b \subseteq \mathcal{A}_a$, guaranteeing $R_{a \rightarrow b} = 1$.

Defining $\kappa_{a \rightarrow b} = \|\delta\|_1 / m_a$, the condition $\kappa_{a \rightarrow b} < 1$ is a sufficient condition for placement stability.

Corollary 1 (Positive-scaling invariance). *If $w^{(b)} = c w^{(a)}$ for all edges in the common edge universe with scalar multiplier $c > 0$, then the ordering of all feasible partitions across $\mathcal{F}$ is strictly preserved. Consequently, the optimal-partition sets coincide, $\mathcal{A}_a = \mathcal{A}_b$, and $R_{a \rightarrow b} = R_{b \rightarrow a} = 1$.*

Proof. For any partition $P \in \mathcal{F}$, the objective evaluates to $J_b(P) = \sum_e w_e^{(b)} z_e(P) = c \sum_e w_e^{(a)} z_e(P) = c J_a(P)$. Because $c > 0$, the transformation is a strictly monotonic order-preserving bijection on the objective values across $\mathcal{F}$. Thus $J_a(P) \leq J_a(Q) \iff J_b(P) \leq J_b(Q)$ for all $P, Q \in \mathcal{F}$, which guarantees $\arg \min_{P \in \mathcal{F}} J_a(P) = \arg \min_{P \in \mathcal{F}} J_b(P)$, i.e., $\mathcal{A}_a = \mathcal{A}_b$, immediately yielding $R_{a \rightarrow b} = R_{b \rightarrow a} = 1$.

D. Set-level evaluation

Fragment labels are symmetric: permuting label names leaves the physical partition and the cut-indicator vector $z(P)$ unchanged. Symmetry breaking can remove these duplicate labelings, but it does not remove distinct optimal partitions with different cut vectors. All representation-regret experiments use one symmetric near-balanced capacity policy,

$$L_k = \left\lfloor \frac{n}{K} \right\rfloor, \quad U_k = \left\lceil \frac{n}{K} \right\rceil \quad (k = 1, \dots, K), \quad (6)$$

for both exhaustive enumeration and SCIP. Thus $K = 2$ is exactly balanced for the even sizes studied, while $K = 3$ admits only the near-balanced load permutations

implied by Eq. (6). We evaluate (5) with two optimization stages. Stage 1 obtains J_a^* and J_b^* . Stage 2 minimizes $J_b(P)$ subject to $J_a(P) \leq J_a^* + \tau_{\text{opt}}$; exchanging a and b gives the reverse direction.

For finite optimality margins, exact set identification requires

$$0 < \tau_{\text{opt}} < \min\{m_a, m_b\}. \quad (7)$$

The numerical implementation also chooses τ_{opt} above the solver feasibility scale and any residual Stage-1 objective gap. We use exhaustive enumeration for $n \leq 10$ and a lexicographic SCIP solve for $n \in \{12, 14, 16\}$. The reported representation conclusions are unchanged for $\tau_{\text{opt}} \in \{10^{-7}, 10^{-8}, 10^{-9}\}$; solver settings and tolerance checks are given in Sec. V A and the archived data.

IV. CAPACITATED FORMULATION AND RESOURCE BOUNDS

A. Capacitated K -way partition problem

Let $K \geq 2$ be the declared fragment count. Binary assignment $a_{ik} = 1$ denotes that qubit i belongs to fragment k , with capacities $L_k \leq \sum_i a_{ik} \leq U_k$. For each aggregated interaction edge $e = (i, j)$, locality variable $y_{ek} = 1$ denotes that both endpoints occupy fragment k , while $z_e = 1$ denotes a crossed interaction. The exact independent-QPD MIP is

$$\begin{aligned} \min \quad & \sum_{e \in E} w_e z_e \\ \text{s.t.} \quad & \sum_{k=1}^K a_{ik} = 1 \quad \forall i, \\ & L_k \leq \sum_i a_{ik} \leq U_k \quad \forall k, \\ & y_{ek} \leq a_{ik}, \quad y_{ek} \leq a_{jk} \quad \forall e = (i, j), k, \\ & y_{ek} \geq a_{ik} + a_{jk} - 1 \quad \forall e = (i, j), k, \\ & z_e + \sum_{k=1}^K y_{ek} = 1 \quad \forall e, \\ & a_{ik}, y_{ek}, z_e \in \{0, 1\}. \end{aligned} \quad (8)$$

The three inequalities are the exact binary AND hull. Since each endpoint has exactly one assignment, exactly one y_{ek} is one iff both endpoints share a fragment; otherwise all locality variables are zero and $z_e = 1$. Thus (8) exactly optimizes (4) for arbitrary feasible capacities.

Proposition 4 (Objective equivalence for capacitated K -way partitioning). *For any $K \geq 2$ and feasible capacities, the objective of (8) equals the independent gate-level log overhead (2).*

Proof. The assignment/locality constraints set $z_e = 1$ exactly when the endpoints of aggregate edge e are in different fragments. Equation (3) sums the log overhead

of every gate occurrence in G_e . Regrouping (2) by edge yields $\sum_e w_e z_e$.

This equivalence formalizes the reduction from independent K -way gate cutting to Capacitated Weighted Graph Partitioning (CWGP) in the logarithmic domain. In graph-theoretic terms, each representation choice induces a distinct edge-weighted interaction graph $G = (V, E, w^{(r)})$ on the shared vertex set V , providing a concrete setting where the underlying optimization instance is governed by representation-dependent graph weightings. Problem (8) is NP-hard because exact-balanced weighted bisection is its $K = 2$ special case.

B. Fixed-capacity cardinality identity

If fragment sizes are fixed exactly as $|F_k| = n_k$, every feasible partition crosses

$$\begin{aligned} \sum_{i < j} \mathbb{I}(f(i) \neq f(j)) &= \binom{n}{2} - \sum_{k=1}^K \binom{n_k}{2} \\ &= \frac{1}{2} \left(n^2 - \sum_{k=1}^K n_k^2 \right). \end{aligned} \quad (9)$$

complete-graph pairs. This is a structural identity, not a constraint for lower/upper-only capacities. Exact-balanced bisection is the special case $K = 2$, $n_1 = n_2 = n/2$, yielding $n^2/4$. This identity, and the polyhedral strengthenings it enables, underlie the $K = 2$ reference implementation and the negative polyhedral ablation reported in Appendix B.

C. Gap-to-overhead resource bounds

Proposition 5 (Gap-to-overhead translation, exact arithmetic). *For any $K \geq 2$, feasible capacity bounds, and optimization backend, if $\text{LB} \leq \text{OPT} \leq \text{UB}$, where $\text{UB} = J(P_{\text{inc}})$ for a feasible incumbent of (8), then*

$$\frac{\Gamma(P_{\text{inc}})}{\Gamma^*} = \exp(\text{UB} - \text{OPT}) \leq \exp(\text{UB} - \text{LB}). \quad (10)$$

Proof. Equation (2) gives $\Gamma(P) = \exp(J(P))$. Since $\text{OPT} \geq \text{LB}$, exponentiation yields (10).

This result depends on the specified independent-QPD objective and on valid global bounds, not on how a backend obtains them. Proposition 6 (Appendix A) establishes the required bound premise for the $K = 2$ reference search in exact arithmetic; mature MIP solvers supply the same premise through their own dual bounds.

D. Numerical interpretation

Proposition 5 is an exact statement about valid lower and upper bounds. Experimental MIP quantities are computed in floating-point arithmetic and are interpreted at the stated solver tolerances. The representation experiments use `numerics/feastol` = 10^{-10} , while the broader scaling experiments use their reported SCIP settings and objective-gap criteria. Appendix G provides a separate small CX-only decimal interval-enclosure check. Full numerical settings are archived with the released data.

V. NUMERICAL RESULTS

A. Computational protocol

Experiments use Python 3.11.9, Qiskit 2.5.1, Qiskit Addon Cutting 0.10.0, Qiskit Aer 0.17.2, SciPy 1.17.1 [13], KaHIP 3.25 [14], MQT Bench 2.2.2 [15], MQT QCEC 3.10.0 [16], PySCIPOpt 6.2.1, and SCIP 10.0.2 [17]. Runs were performed on Windows 11 with an Intel Core i5-14400 and 31.69 GB RAM, with numerical libraries pinned to one thread. Raw records, seeds, solver settings, software versions, and SHA-256 manifests are included in the revision-specific release package described in the Data availability statement; the cited Zenodo v1.0.0 record corresponds to the prior manuscript version. Reported proportions summarize this benchmark design and are not population estimates.

Generative AI tools Claude (Anthropic, Opus 4.8 model family) and ChatGPT (OpenAI, GPT-5.6 Sol model family) assisted with implementation and debugging of experiment and analysis code. The authors directed the tasks, executed the resulting code, and checked the relevant outputs against exhaustive enumeration, solver-consistency tests, and numerical regression tests. Scientific claims, derivations, and conclusions remain the responsibility of the authors.

B. Representation construction and semantic checks

Each pair is generated from the same logical source circuit and retains the same logical wire set V without ancillas. Any symbolic source parameters are bound to the fixed value $\pi/4$ before either representation is constructed; the study therefore evaluates fixed benchmark instances rather than a parameter sweep. The CX-normalized representation is produced with Qiskit's basis $\{\text{rz}, \text{sx}, \text{x}, \text{cx}\}$, optimization level 1, and transpiler seed 0. The native-QPD representation recursively expands unsupported composite operations while retaining supported native two-qubit gates. Both constructions use `coupling_map=None`, so no routing transformation is introduced.

The source fingerprint is parameter-sensitive: it hashes the ordered instruction names, wire indices, numeric parameters, circuit dimensions, and global phase. Semantic checks are then performed independently of that provenance record where computationally practical. Small pairs are compared as Qiskit `Operator` matrices up to global phase. For the medium QAOA, QFT, and exact-QPE pairs at $n \in \{12, 14, 16\}$, MQT QCEC 3.10.0 verifies equivalence up to global phase using its equivalence-checking pipeline [16]; this includes the $n = 14$ QPE main witness. The $n = 10$ Grover exact-placement witness is checked by applying both circuits to deterministic Haar-random statevectors and comparing the outputs up to global phase; this fixed-seed check is probabilistic rather than a formal proof. For the deepest medium Grover pairs, the release records instead retain construction provenance: the two representations originate from the same parameter-sensitive source fingerprint and wire order, routing and dynamic operations are absent, the native representation recursively expands operation definitions, and the CX representation uses Qiskit’s semantics-preserving basis translation. The result records state the verification method used for every pair.

C. Representation-induced placement reversals

Table I summarizes 96 paired representation records. Three exact-QPE records have disjoint optimal-placement sets and independent semantic checks. The clearest witness occurs at $n = 14$, $K = 2$, where $\Delta_{\text{native} \rightarrow \text{CX}} = 6.5917$ and $R = 729$. Although this record is also solved by the medium-tier SCIP pipeline, an independent exhaustive enumeration of all 1,716 balanced bipartitions gives the same objectives and regrets, with one optimum in each representation and no overlap between the argmin sets. The $n = 14$, $K = 3$ QPE change is also reproduced by exhaustive enumeration of 126,126 near-balanced partitions, and the remaining QPE change is already in the exact $n = 6$, $K = 2$ tier. Thus all three QPE changes have exhaustive support. Five additional highly expanded Grover records have positive set-level regret at the stated solver tolerances under the recorded construction provenance; because those representations are much larger and the deepest pairs are not independently equivalence-checked, we treat them as supporting stress cases in Appendix D rather than as the primary evidence. The SCIP conclusions are unchanged across the tested τ_{opt} values.

For the fixed- $\pi/4$ QAOA benchmark instances, the CX-normalized interaction weights are a uniform positive rescaling of the native RZZ weights, so Corollary 1 predicts the observed $R = 1$ for every tested size. Among exhaustive records satisfying $\kappa < 1$, all 25 also satisfy $R = 1$ as required by Proposition 3. The extreme Grover cases are reported separately in Appendix D because their representation expansion produces overhead scales far outside executable regimes.

D. Elimination of false-positive QFT reversals

On the reference SCIP run, comparing one returned solution from each representation produced six apparent QFT placement changes, with a largest cross-evaluated ratio of 27,741. The two-stage calculation gives $\Delta = 0$ and $R = 1$ in all six cases. We independently exhaustively enumerated the same six feasible spaces at $n \in \{12, 14, 16\}$ and $K \in \{2, 3\}$, including 1,009,008 near-balanced partitions for $n = 16$, $K = 3$; every case has a nonempty intersection between the CX and native argmin sets. Thus the apparent changes are produced by the particular degenerate optimum returned by each single solve, rather than by disjoint optimal-placement sets.

VI. RELATED WORK

Circuit cutting based on quasiprobability decompositions traces to work on simulating larger circuits with smaller quantum devices and on virtual nonlocal gates [2, 3, 12, 18]. CutQC couples partition selection with classical recombination [4], and Qiskit Addon Cutting provides gate-level QPD experiment generation and reconstruction [11]. Sampling and reconstruction overhead have subsequently been reduced through improved QPDs, randomized measurements, joint decompositions, classical communication, commuting-group wire cuts, online pruning of reconstruction terms, and overhead-constrained knitting [19–32]. Hart [33] further characterizes optimal two-qubit gate-cutting cost through nonlocality descriptors, reinforcing that gate-cutting cost itself is gate dependent.

Automatic cut-placement methods typically optimize the cut choice for a fixed circuit. Brandhofer *et al.* [8] optimize sampling-aware gate and wire cuts, Nakamura *et al.* [9] study scalable multi-fragment partitioning and sampling bounds, and Idan *et al.* [34] analyze circuit-cutting complexity and combinatorial optimization. ScaleQC and FitCut address scalable automatic cutting and distributed execution [35, 36]; Circuit Folding exploits modular graph structure [37]; Fröhler *et al.* consider gate, wire, and joint gate-group cuts [10]; and Eslaminia and Le Beux jointly optimize cut count and routed-subcircuit depth [38]. Hardware-aware methods such as MOSAIQC and TW2S incorporate mixed cut types, routing, hardware structure, or noise non-uniformity [39–41]. CutReg makes sampling overhead an adaptive regularizer during QML optimization [42]. These studies make clear that cut placement and cutting overhead are already active optimization targets; our contribution is not a new general-purpose placement algorithm.

Decomposition-aware cutting also predates the present study. Lian *et al.* [43] use virtual two-qubit decomposition to reduce the minimum cut count in QFT circuits, while Tamura *et al.* [44] choose multi-qubit decompo-

TABLE I. Cross-representation placement regret across 96 paired records (Δ in natural-log overhead units). Records with $n \leq 10$ are exhaustive; larger records use the two-stage SCIP evaluation of Sec. IIID. Independent exhaustive audits additionally cover both $n = 14$ QPE records and all six QFT records at $n \in \{12, 14, 16\}$, $K \in \{2, 3\}$. Row-level SCIP conclusions are unchanged across the tested τ_{opt} values. Extreme Grover magnitudes are deferred to Appendix D.

Family	Records	Set-level changes	Max $\Delta_{\text{CX} \rightarrow \text{native}}$	Max $\Delta_{\text{native} \rightarrow \text{CX}}$
QAOA	12	0	0.0	0.0
QFT	12	0	0.0	0.0
QPE-exact	12	3 (25%)	3.31	6.59 ($\log_{10} R = 2.86$)
Grover	12	5 (42%)	Appendix D	Appendix D
BV	12	0	0.0	0.0
Controls (VQE, GHZ, DJ)	36	0	0.0	0.0

sitions conditioned on a selected partition boundary to lower sampling overhead. These works establish that decomposition can materially affect a cutting strategy; they do not evaluate whether two complete equivalent representations induce different QPD-weighted argmin sets over the same feasible partition maps. Outside circuit cutting, semantics-preserving circuit rewrites have also been used to reduce inter-QPU communication cost in distributed compilation [45]. Our narrower question fixes the logical wire set, fragment count, capacity family, and independent-QPD resource model, treats two semantics-preserving complete gate-level representations as alternative inputs, and compares their full optimal-placement sets so that optimizer tie breaking cannot create a false change. We therefore do not claim the first use of decomposition to improve circuit cutting, nor a new generic graph-partitioning algorithm; the optimization machinery is standard [5–7].

VII. DISCUSSION AND LIMITATIONS

The QPE and QFT results show complementary sides of representation sensitivity: a representation change can move the optimal-placement set, while a comparison based on one returned optimum can also create a false change when the optimum is degenerate. The central object is therefore the argmin set under a fixed feasible partition space and resource model; the set-level cross-evaluation distinguishes a genuine representation-induced change in that set from solver selection among degenerate optima. Proposition 2, Proposition 3, and Corollary 1 give simple perturbation diagnostics that bound or exclude such changes. These inequalities are elementary sensitivity consequences rather than a claim of new generic optimization theory; their purpose here is to make the representation effect testable for the circuit-cutting objective.

Changing the gate list can change absolute QPD cost without changing the minimizing partition. The controls make this distinction explicit: QAOA preserves the argmin set because its interaction weights rescale uniformly, while QFT changes the weight pattern substantially yet retains at least one shared optimum in every tested medium case. The QPE examples therefore iden-

tify decision-level representation dependence rather than merely a change in numerical overhead.

The study is confined to gate cuts under the independent per-gate QPD model on a fixed logical wire set and fixed capacity constraints. Wire cuts, routing, hardware noise, and general joint decompositions are not part of that primary optimization problem; the restricted joint experiment in Appendix F illustrates that changing the resource model can also change the preferred placement. Numerical MIP results use the stated solver tolerances, while Appendix G provides a separate decimal interval-enclosure sanity check on a small CX-only calculation. The benchmark collections are fixed test sets, so reported frequencies describe those sets rather than a workload distribution.

VIII. CONCLUSION

Semantically equivalent gate representations can induce different resource-optimal cut placements under an independent-QPD cost model. The set-level regret in Eq. (5) separates such changes from degeneracy in the optimizer’s returned solution: the QPE benchmark gives a genuine $R = 729$ change, whereas six apparent QFT changes reduce to $R = 1$ under the set-level cross-evaluation. The perturbation and stability results quantify when representation changes can or cannot alter that decision. For a fixed representation, the same QPD objective is exactly a capacitated weighted graph-partitioning problem in the log domain, with global optimization gaps translating directly into multiplicative modeled-overhead bounds. Circuit representation should therefore be treated as part of the resource model when evaluating or optimizing cut placement.

ACKNOWLEDGMENTS

The authors thank the maintainers of the open-source tools used in this work (Qiskit, Qiskit Addon Cutting, MQT Bench, KaHIP, and SCIP) for enabling reproducible experimentation. Generative AI tools Claude (Anthropic, Opus 4.8 model family) and ChatGPT (OpenAI, GPT-5.6 Sol model family) were used for language

editing, structural revision, and preparation of schematic figure layouts. The authors retain responsibility for the scientific content and final manuscript.

AUTHOR CONTRIBUTIONS

Phung Trong Hung: conceptualization, methodology, software, formal analysis, investigation, visualization, and writing—original draft. Huong Bui: supervision and writing—review and editing. Both authors reviewed and approved the final manuscript and take responsibility for its content.

DATA AVAILABILITY

A prior version of the software and data is archived at Zenodo [46]. A revision-specific archive containing the corrected canonical representation data, exhaustive audits, regression tests, and integrity manifest has been prepared for deposition as a new Zenodo version. The final resubmission will cite that version-specific record; the release manifest records the source revision and SHA-256 hashes of the manuscript-critical files.

Appendix A: Optimization backends and reference implementation

This appendix records the optimization backends behind the results of Sec. V. The scientific claims of the paper do not depend on which backend supplies the valid bounds required by Proposition 5; we retain a specialized reference search only as an independent algorithmic reference and, on the evaluated corpus, an off-the-shelf MIP solver outperforms it.

1. Validation oracles and relaxations

We use a compact MIP and brute-force enumeration as independent validation oracles. For one pair (i, j) , the convex hull of $x_{ij} = |z_i - z_j|$ is described by

$$\begin{aligned} x_{ij} &\geq z_i - z_j, & x_{ij} &\geq z_j - z_i, \\ x_{ij} &\leq z_i + z_j, & x_{ij} &\leq 2 - z_i - z_j. \end{aligned} \quad (\text{A1})$$

The implementation uses an equivalent one-hot assignment model whose MIP retains only the two lower inequalities in (A1) and declares x_{ij} binary; for positive-weight interacting pairs, minimization sets x_{ij} to its exact XOR value. SciPy/HiGHS solves this MIP, while exhaustive enumeration through 16 qubits supplies a second oracle. Neither oracle contributes bounds to the anytime results. The basic reference LP (A0) uses one-hot assignment variables and cut variables only for interacting

pairs, retains exact balance and the two lower XOR inequalities from (A1), and relaxes all variables to $[0, 1]$; because the weights are nonnegative, this is a valid but weak lower relaxation.

2. Exact balanced $K = 2$ reference problem

The specialized reference implementation uses the $K = 2$ exact-balanced special case. All legacy reference circuits have even n ; the reference requires $n/2$ qubits per fragment and fixes $z_0 = 0$ to remove label symmetry:

$$\min_{z \in \{0,1\}^n} \sum_{i < j} w_{ij} |z_i - z_j| \quad \text{s.t.} \quad \sum_i z_i = n/2, \quad z_0 = 0. \quad (\text{A2})$$

Every feasible bisection cuts exactly $(n/2)^2$ pairs in the complete-pair representation, $\sum_{i < j} x_{ij} = n^2/4$, the special case of (9) used by the strengthened relaxation of Appendix B.

3. Reference branch and bound and its safe boundary

The reference optimizer maintains open nodes in non-decreasing order of their LP bounds, expands the best-bound node, fathoms integral nodes after updating the incumbent whenever it improves UB, and otherwise branches on the most fractional unfixed label. Child LPs reuse the root cut pool of Appendix B; any child whose lower bound is at least UB is pruned. After all LP solves associated with an expansion complete, the algorithm reaches a safe boundary and computes

$$\text{LB} = \min \left(\text{UB}, \min_{N \in \mathcal{N}_{\text{open}}} \text{LB}_N \right), \quad (\text{A3})$$

with $\min_{N \in \emptyset} \text{LB}_N = +\infty$, so that $\text{LB} = \text{UB}$ when the frontier is empty. The additive log-domain gap and multiplicative overhead factor are $\Delta = \text{UB} - \text{LB}$ and $F = \exp(\Delta)$.

Proposition 6 (Reference safe-boundary validity, exact arithmetic). *In exact arithmetic, every reference-search safe stopping boundary satisfies $\text{LB} \leq \text{OPT} \leq \text{UB}$.*

Proof. Feasibility of the incumbent gives $\text{OPT} \leq \text{UB}$. Every unexplored feasible solution belongs to a frontier subtree whose LP value lower-bounds all integral completions in that subtree. Equation (A3) therefore gives $\text{LB} \leq \text{OPT}$.

Experimental deadlines are soft: the reference optimizer stops only at a safe boundary, and we report the actual boundary time. Together with Proposition 5, Proposition 6 supplies the exact-arithmetic premise for the anytime resource bound $\Gamma_{\text{inc}}/\Gamma^* \leq \exp(\text{UB} - \text{LB})$; in practice every reported bound uses floating-point HiGHS or SCIP

TABLE II. Heterogeneous-QPD backend comparison. Entries report counts with $F \leq 1.01$, the reporting threshold; on this corpus this set coincides with the runs meeting the declared solver-tolerance objective-gap closure criterion (10^{-9}). All factors and closures are solver-tolerance quantities.

Method	2 s	10 s	60 s	60 s UB	match G0
Reference	160/200	184/200	199/200		199/200
SCIP G0	199/200	200/200	200/200		reference
SCIP G1	111/200	186/200	196/200		198/200
SCIP G2	108/200	158/200	183/200		189/200

output and is a solver-tolerance quantity (Sec. IV D). Figure 2 states the full anytime search. Two greedy incumbents seed the search: H0 assigns vertices by descending weighted-degree greedy insertion with first-improving single swaps, while H2 adds a spectral (Fiedler) initialization and best-improving Kernighan–Lin swaps; neither establishes a lower bound.

4. Mature MIP is the strongest evaluated backend

Three SCIP variants differ in formulation strengthening: G0 is the sparse basic model (cut variables only for interacting edges and the two lower XOR inequalities, no cardinality); G1 augments G0 with complete-pair cut variables, the full XOR hull, and the cardinality identity $\sum x_{ij} = (n/2)^2$; and G2 further adds the full triangle/metric inequalities (B1)–(B4). Table II reports all 200 heterogeneous-QPD instances at each independent checkpoint. SCIP G0 is the strongest practical backend among the evaluated methods, reaching solver-tolerance closure on 199/200 instances at 2 s and all 200 by 10 s; the specialized reference implementation closes 160/200, 184/200, and 199/200 at 2, 10, and 60 s. At 60 s the reference matches the G0 incumbent on 199/200 records, its single open case being a dense-shuffled $n = 40$ instance with $F = 6.1262$. Statically adding cardinality or full triangles to SCIP does not improve its end-to-end performance on this corpus. These results do not support a specialized-solver performance advantage; they show that the log-domain model-factor semantics can be instantiated by either the reference implementation or a mature MIP solver. All methods run single-threaded and, at each independent wall-clock checkpoint, emit common UB, LB, and $F = \exp(\text{UB} - \text{LB})$ fields, which are solver-tolerance quantities.

Appendix B: Polyhedral formulations and negative ablations

This appendix records the $K = 2$ polyhedral strengthenings and the negative ablations that motivated the sparse production formulation of Sec. IV. They are supporting context, not additional research claims.

1. Balanced-cut cardinality and metric inequalities

The B2S reference lifts A0 to the complete pair space, imposes the full XOR hull (A1), and adds two families of valid constraints. Balanced-cut cardinality: the identity $\sum_{i < j} x_{ij} = n^2/4$ from (A2) fixes the total cut mass. Metric inequalities: for each triple (i, j, k) , the cut polytope satisfies [5]

$$x_{ij} - x_{ik} - x_{jk} \leq 0, \quad (\text{B1})$$

$$x_{ik} - x_{ij} - x_{jk} \leq 0, \quad (\text{B2})$$

$$x_{jk} - x_{ij} - x_{ik} \leq 0, \quad (\text{B3})$$

$$x_{ij} + x_{ik} + x_{jk} \leq 2. \quad (\text{B4})$$

Inequalities (B1)–(B3) are valid facets of the multiway-cut polytope for arbitrary K , whereas (B4) is specific to the $K = 2$ cut polytope and is imposed only in the $K = 2$ formulations: for $K \geq 3$, three qubits may occupy three distinct fragments, making $x_{ij} + x_{ik} + x_{jk} = 3$ feasible. B2S solves the root LP, separates violated inequalities until none remain or the separation budget is exhausted, and reuses the resulting pool throughout the tree with no node-level separation. Let CP0 denote the complete-pair LP relaxation carrying the full pairwise XOR hull (A1) but neither the balanced-cardinality identity nor the metric inequalities (B1)–(B4).

Proposition 7 (Objective-bound redundancy without cardinality). *For nonnegative weights, adding (B1)–(B4) to the complete-pair baseline CP0 does not change its optimal value. This is an objective statement, not a claim that the metric inequalities leave the CP0 polyhedron unchanged.*

Proof. Take an optimal CP0 solution and retain its label vector z . Set $x_{ij} = |z_i - z_j|$ for every pair. This extension satisfies the pairwise XOR hull and all metric inequalities because absolute distance on the line is a metric and, for $z_i \in [0, 1]$, $|z_i - z_j| + |z_i - z_k| + |z_j - z_k| \leq 2$. For fixed z , these are the smallest feasible pair values; nonnegative weights therefore ensure that the objective does not increase, so the metric-augmented optimum is no larger than the CP0 optimum. The reverse inequality follows because constraints were added.

Proposition 8 (Relaxation validity). *The B2S LP optimum is a valid lower bound on the $K = 2$ reference OPT.*

Proof. For binary labels, $x_{ij} = |z_i - z_j|$ satisfies (A1). A balanced partition cuts $|F_0||F_1| = n^2/4$ complete-graph pairs, and each label triple satisfies (B1)–(B4). Hence every feasible integral bisection remains feasible in the relaxation, whose minimum cannot exceed OPT.

Proposition 9 (Trivial sparse root relaxation). *For exact capacities $L_k = U_k = n_k$, the LP relaxation of the sparse edge-only formulation (8) has objective zero.*

Algorithm: Reference best-bound search	
Input:	QPD-weighted interaction graph G , capacity $q_{\max} = n/2$, soft deadline T .
Step 1.	Compute initial feasible partition and upper bound: $(P_{\text{inc}}, \text{UB}) \leftarrow \text{H2}(G)$.
Step 2.	Solve and strengthen root LP via separation: $(\text{LB}_0, \mathcal{C}) \leftarrow \text{B2S-ROOT}(G)$. Fix cut pool $\mathcal{C}$.
Step 3.	Initialize search frontier $\mathcal{N}_{\text{open}}$ with the root node.
Step 4.	While $\mathcal{N}_{\text{open}} \neq \emptyset$ and elapsed time $< T$:
•	Select the open node N^* of least lower bound LB_{N^*} .
•	If N^* is integral, update the incumbent when it improves UB, then fathom N^* .
•	Otherwise, branch on the most fractional unfixed qubit; solve child LPs using $\mathcal{C}$; prune children with $\text{LB}_N \geq \text{UB}$.
Step 5.	At each safe boundary, compute the global lower bound (A3) and emit a bound report.
Output:	Best partition P_{inc} and bound report $(\text{LB}, \text{UB}, F = \exp(\text{UB} - \text{LB}))$.

FIG. 2. Reference best-bound search for exact-balanced $K=2$ partitioning.

Proof. Set $a_{ik} = n_k/n$ and $y_{ek} = n_k/n$ for every qubit i , edge e , and fragment k , then set $z_e = 0$. Assignment and capacity equalities hold since $\sum_k n_k = n$. The upper AND inequalities hold at equality; the lower inequality holds because $n_k/n \geq 2n_k/n - 1$. Finally $z_e + \sum_k y_{ek} = 1$. Thus this is LP feasible with zero nonnegative objective.

2. Negative ablations

Motivated by Proposition 9, we tested an all-pair extension of (8) that imposes (9) with auxiliary nonedge variables and complete-metric triangles. This is a negative ablation, not a proposed production formulation: for $K > 2$, materializing all pair variables and triangle constraints expands the representation to $\Theta(n^2)$ pairs and $\Theta(n^3)$ triples, increasing LP construction, memory traffic, and reoptimization cost at every node, while adding only the multiway-cut-valid triangles (B1)–(B3). On a 36-record $n \in \{20, 32\}$ ablation at 5 s, the base formulation closes 31 records, symmetry-only closes 29, and the cardinality or cardinality-plus-metric variants close 6 each. Production runs therefore use SCIP’s sparse assignment/locality formulation with only the valid equal-capacity anchor.

Table III decomposes the root relaxation on 24 matched complete-pair instances: cardinality (C) and metric (T) inequalities become jointly effective only when combined (12/24 roots closed under CT versus 0/24 under CP0, C, or T alone), consistent with Proposition 7. Table IV confirms B2S reference-search closure under a 100-node budget; it does not establish a production advantage over SCIP. Table V breaks down the largest legacy CNOT-only reference size ($n = 40$) by graph family.

TABLE III. Root LP gap and time across cut-pool combinations ($n = 24$, legacy CNOT instances). The full formulation (CP0 + C + T) closes the gap on 12/24 instances at root.

Formulation	Closed	Med. gap	Max gap	Mean time
A0	0/24	26.2935	52.4468	0.0215 s
CP0	0/24	26.2935	52.4468	0.2882 s
CP0 + C	0/24	26.2935	52.4468	0.3429 s
CP0 + T	0/24	26.2935	52.4468	0.2832 s
CP0 + C + T	12/24	0.2848	3.8761	2.3222 s

TABLE IV. Controlled ablation on 90 synthetic balanced instances under a 100-node budget. “Closed” means solver-tolerance gap at most 10^{-9} . “Node-lim.” is a non-exclusive count.

Config.	Closed	Node-lim.	Nodes	Root/Tree (s)
A0 + H0	41/90	56	100	0.000 / 2.059
A0 + H2	41/90	52	100	0.000 / 2.072
B2S + H0	90/90	0	1	0.260 / 0.060
B2S + H2	90/90	0	1	0.256 / 0.063

TABLE V. Family-level performance breakdown at $n = 40$ for the legacy CNOT-only reference. Each family contains ten seeded instances.

Family	Closed	Med. time (s)	p90 time (s)
Community	10/10	0.809	1.186
Nearest-neighbor	10/10	1.646	1.764
QAOA ring	10/10	5.367	6.173
Noisy-community	9/10	2.419	18.188
Random	10/10	18.344	41.224
Dense	10/10	24.070	52.669
Weighted-random	8/10	37.215	60.067

Appendix C: Gate-dependent costs versus cut count

1. Controlled and algorithm-derived examples

Unweighted cut count can rank feasible plans incorrectly. Two iSWAP cuts have modeled overhead $49^2 = 2,401$, whereas three CNOT cuts cost $9^3 = 729$; the

TABLE VI. Algorithm-derived MQT Bench audit. “Max factor” is descriptive over six fixed sizes per family, not a population estimate.

Family	Eligible	Reversals	Max factor
Draper QFT adder	6/6	4/6	1.04×10^6
HHL	6/6	0/6	1
Exact QPE	6/6	2/6	4.03
Inexact QPE	6/6	0/6	1
QFT	6/6	0/6	1
Entangled QFT	6/6	0/6	1

three-cut plan is cheaper. Likewise, two CS cuts cost $(3 + 2\sqrt{2})^2 \approx 33.97$, compared with 81 for two CNOT cuts. Equation (3) preserves these gate-dependent distinctions.

We test this distinction under controlled heterogeneity on 120 deterministic circuits (five families, $n \in \{12, 14, 16\}$) using validated Qiskit costs for CX, CZ, iSWAP, and three nonzero RZZ angles, with no unit-overhead operation. To avoid tie-breaking artifacts, we compare $\min_{P: c(P)=c^*} J(P)$, the best QPD objective among all minimum-cut partitions, with $\min_P J(P)$; both values are obtained by exhaustive enumeration. This stronger regret is strict in 16/120 cases (13.3%), with conditional factor median 5.30, 90th percentile 27.47, and maximum 181.11. Every strict reversal requires more cuts under a QPD-optimal plan (median count increase one, maximum three). Across the 16 reversals, QPD-optimal plans cut 28 fewer iSWAP gates while accepting 47 additional lower-overhead gates.

We then test whether this mechanism occurs without constructed gate palettes. The algorithm-derived study uses six MQT Bench families [15], six even sizes, the eligibility criterion, and the decision rule before inspecting outcomes; it selectively expands only operations with arity above two, preserves numeric one- and two-qubit semantic operations, never transpiles to a target basis, and queries every retained two-qubit occurrence via `QPDBasis.from_instruction`. The count objective $c(P)$ and QPD objective $J(P)$ define the set-level regret

$$\Delta_{E6} = \min_{P: c(P)=c^*} J(P) - \min_P J(P), \quad c^* = \min_P c(P), \quad (C1)$$

and a strict reversal requires $\Delta_{E6} > 10^{-10}$. Fixing qubit zero removes complementary-label symmetry, leaving $\binom{n-1}{n/2-1}$ balanced partitions, all of which are exhaustively evaluated (from 10 at $n = 6$ through 6,435 at $n = 16$). All 36 circuits are eligible: none has an unsupported or unit-overhead two-qubit operation, and each has at least two positive QPD-cost classes. Exhaustive optimal-set enumeration finds six strict reversals: four among six Draper QFT adders and two among six exact-QPE circuits (Table VI). HHL, inexact QPE, QFT, and entangled QFT supply 24 negative results despite heterogeneous costs. Thus heterogeneity is necessary for the observed mechanism but is not sufficient to change the optimum.

Draper QFT adder ($n = 16$): cut topology under count vs. independent-QPD objectives

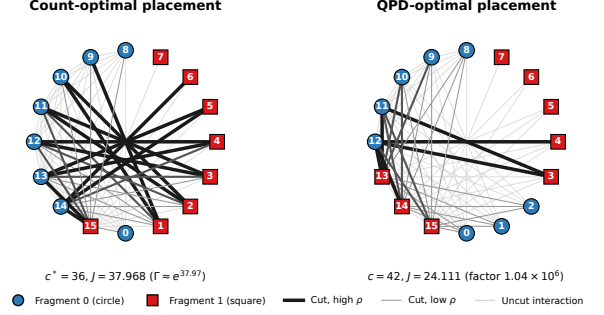


FIG. 3. Draper QFT-adder interaction topology at $n = 16$ under exact balanced bipartition. Left: count-optimal placement ($c^* = 36$, $J \approx 37.97$). Right: independent-QPD-optimal placement ($c = 42$, $J \approx 24.11$). Darker cut edges carry larger $\log \rho$; the QPD plan trades cut count for avoidance of high-overhead controlled phases.

TABLE VII. Grouped nonzero-delta gate-class accounting for the Draper $n = 16$ witness. The groups sum to +6 cuts and $J_{\text{QPD}}^* - J_{\text{count}}^* = -13.856805$; full per-angle accounting is in the reproducibility artifact.

CP angle class	Count	QPD	Δc
$-\pi/64$ through $-\pi/4$	5	13	+8
$\pi/64$ through $\pi/16$	6	15	+9
$\pi/4$	5	2	-3
$\pi/2$	6	2	-4
π	6	2	-4
Total	28	34	+6

The largest observed witness is the Draper QFT adder at $n = 16$: the best count-optimal plan has 36 cuts and $J = 37.968133$, whereas the QPD-optimal plan has 42 cuts and $J = 24.111328$, yielding a factor 1.042×10^6 . The QPD plan avoids expensive $CP(\pi/2)$ and $CP(\pi)$ crossings by accepting more small-angle controlled phases (Table VII). Direct occurrence-level sums and independently aggregated pair weights agree within 10^{-10} for both plans in every reversal record. Figure 3 makes the same Draper $n = 16$ witness geometric: node colors encode the two fragments, while edge darkness encodes independent-QPD log weight; the count-optimal map severs fewer interactions but retains several high- ρ controlled phases across the cut, whereas the QPD-optimal map accepts six additional crossings yet eliminates most expensive phase crossings. This is existence evidence on algorithm-derived benchmark circuits, not a prevalence estimate over workloads or a hardware-routing result.

Appendix D: Extreme Grover representation cases

Five Grover records have positive set-level regret at the stated solver tolerances under the two representations.

These cases also exhibit very large representation expansion and correspondingly large modeled log-overheads, so they are not used as operational examples in the main text. The exact $n = 10$, $K = 2$ case has $\Delta_{\text{CX} \rightarrow \text{native}} = 448.23$ and $\Delta_{\text{native} \rightarrow \text{CX}} = 1494.11$; multiplicative factors are therefore reported in the log domain when direct exponentiation exceeds floating-point range. The largest case, at $n = 16$ and $K = 3$, has $\Delta_{\text{CX} \rightarrow \text{native}} = 9984.19$ ($\log_{10} R = 4336.1$), with 234,300 and 395,044 two-qubit gate occurrences in the CX-normalized and native representations, respectively. The corresponding optimum log-objectives are $J_a^* = 240244.5$ and $J_b^* = 394375.4$, while $\|\delta\|_1 = 371287.2$. The regret is therefore about 2.7% of the total interaction-weight perturbation. We report $\log_{10} R = \Delta / \ln 10$ directly for these cases to avoid numerical overflow. The $n = 16$, $K = 2$ Grover record is also present and has $R = 1$ under the set-level comparison.

Appendix E: Additional optimization scaling results

1. Capacitated K -way heterogeneous scaling

We evaluate the core formulation (8) beyond bisection on a heterogeneous-QPD benchmark of 144 runs (three fixed seeds per family-size- K cell, $n \leq 60$, $K \leq 5$, 10-s SCIP limit). Table VIII and Fig. 4(a) summarize all runs, including open cases. SCIP reaches solver-tolerance closure on 101/144 runs. The boundary is strongly graph- and seed-dependent: community matching and weighted repeat close all three seeds through $(K, n) = (3, 60)$ and $(4, 40)$, whereas random matching is the hard family—all three $K = 2, n = 60$ runs close, but every $K \geq 3, n \geq 40$ run remains open.

Open cases can have very weak multiplicative bounds at the 10-s deadline. For random matching, the median open factor is $10^{15.47}$ at $n = 32, K = 5$ and $10^{43.48}$ at $n = 60, K = 4$. We use these runs only to characterize the computational boundary of this supporting benchmark. A 24-cell seed-0 sensitivity rerun at `numerics/feastol` = 10^{-9} preserves every closed/open classification from the original `feastol` = 10^{-6} runs and reproduces the jointly closed objectives at the recorded precision. Figure 4(b) shows the corresponding fragment-width/overhead tradeoff: larger K reduces maximum fragment width while generally introducing more independent-QPD boundaries.

We further confirm that cut count is a poor proxy for sampling overhead at the placement level using matched-objective heuristics that solve the same fixed independent gate-QPD, exact-capacity K -way problem: greedy swap, a count-weighted pairwise Kernighan-Lin (KL) adaptation, QPD-weighted KL, and KaHIP-QPD with exact-capacity repair [14, 47]. One canonical evaluator recomputes J , fragment sizes, and cut occurrences for every method, and also re-evaluates each reported SCIP incumbent after decoding it to a discrete partition. On

the 101 solver-tolerance-closed records (Table IX), KL-QPD matches the SCIP objective on 68 records with median $\log_{10} R = 0$ and p90 2.94, whereas count-weighted KL matches only 23 with median $\log_{10} R = 1.56$; the two differ only in objective semantics within one heuristic family, isolating the effect of optimizing overhead rather than count. All 144 partitions from every matched method satisfy exact capacities. We also apply the same native independent-QPD model to 36 algorithm-derived MQT Bench circuits: SCIP closes 15 within 10 s, KL-QPD matches all 15 solver-tolerance-closed objectives, and the six Draper-adder records form the largest structured positive subset. Three VQE candidates are retained as explicit ingestion failures because their native two-qubit operations lack a Qiskit QPD basis. Thus the heuristic finding is not solely a synthetic-graph artifact.

Appendix F: Resource-model sensitivity and finite-shot tests

1. Sensitivity to the quasiprobability resource model

We also test dependence on the resource model itself. Under a restricted library of pairwise-disjoint same-layer gate groups, a controlled six-qubit instance makes the independent-QPD-optimal placement $1.90496\times$ more expensive than the restricted-joint optimum when both are evaluated with the restricted parallel-joint cost. This establishes an existence example in which changing the decomposition model changes the preferred placement. General joint-QPD-aware partitioning, wire cuts, and hybrid gate/wire models remain outside the primary model studied here [22, 23, 26].

2. Finite-shot scope

A final study uses equal ideal-Aer shot budgets on three $n = 12$ instances. The $719.66\times$ modeled reversal reduces RMSE for both tested observables, the $3.27\times$ case gives mixed RMSE changes, and the no-reversal control is identical under shared seeds. These runs provide a limited operational check rather than a calibration of finite-shot RMSE from the independent-QPD objective; the archived records include the witness identities and bootstrap intervals.

Appendix G: Decimal interval-enclosure and reconstruction checks

As a numerical cross-check separate from the SCIP runs, we evaluate small CX-only log-overheads with 80-digit directed decimal rounding. With $\rho = 9$ represented exactly, the resulting intervals enclose a higher-precision Decimal evaluation of the same expression. This check

TABLE VIII. Capacitated K -way heterogeneous-QPD scaling, three fixed seeds per cell and a 10-s SCIP limit. “Closed” means solver-tolerance closure. Open records retain finite factors, including $10^{43.48}$ at random-matching $n = 60, K = 4$; this fixed matrix is not a workload-prevalence estimate.

Family	Closed	All-seed closure frontier	Boundary
Random matching	24/48	$K = 2, n = 60$; $K = 4, n = 20$	$K \geq 3, n \geq 40$
Community matching	37/48	$K = 3, n = 60$; $K = 4, n = 40$	$K = 5, n \geq 32$
Weighted repeat	40/48	$K = 3, n = 60$; $K = 4, n = 40$	$K \geq 4, n = 60$
Total	101/144	$n = 60, K = 3$	family/seed dependent

TABLE IX. Matched-objective baseline results. Regret statistics use 101 solver-tolerance-closed records only. A method matches on a record when $|\log_{10} R| \leq 10^{-10}$. KaHIP time includes an unbounded native call and is not budget matched ($\dagger$).

Method	Feasible	Match / closed	Med. ΔJ	Med. $\log_{10} R$	p90 $\log_{10} R$	Med. time
Greedy swap	144/144	23/101	3.33	1.45	10.52	0.100 s
Count-KL	144/144	23/101	3.59	1.56	5.01	0.100 s
KL-QPD	144/144	68/101	0.00	0.00	2.94	0.100 s
KaHIP+repair	144/144	51/101	0.00	0.00	1.47	5.89 s †

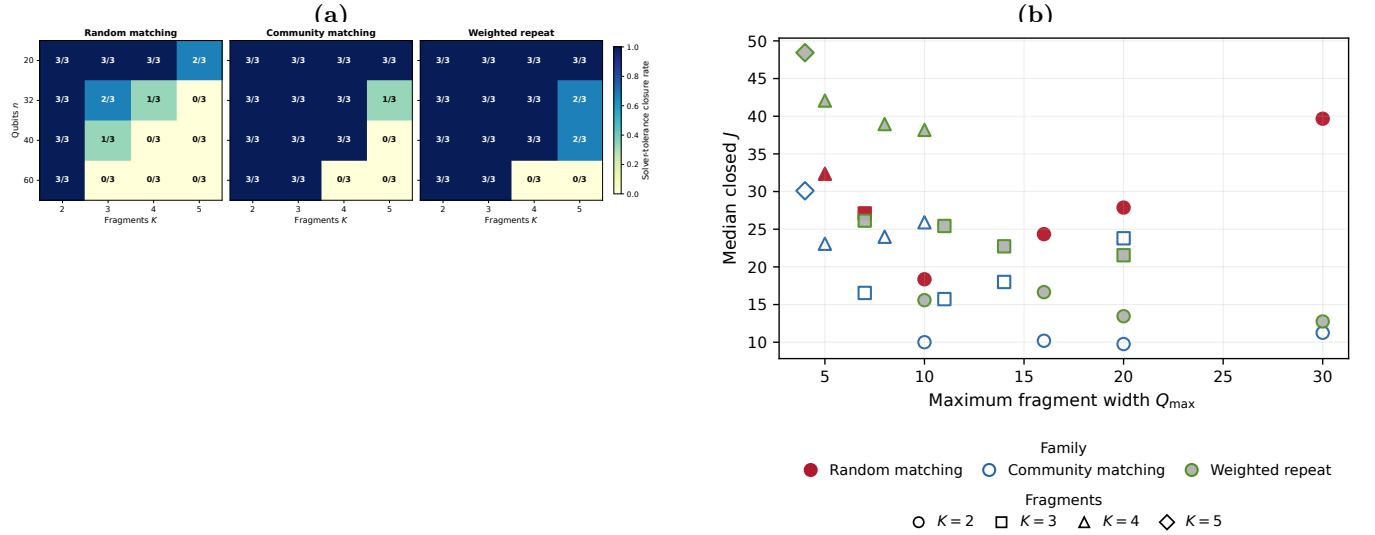


FIG. 4. Supporting K -way scaling views. (a) Solver-tolerance closure counts under a 10-s SCIP limit; each heatmap cell reports closed runs out of three fixed seeds. (b) Width-overhead tradeoff restricted to cells closed for all three seeds; marker and fill conventions distinguish K and graph family. These plots summarize the supporting scaling study and are not used in the representation-sensitivity claims.

TABLE X. Statevector reconstruction check. Predicted and Qiskit-derived overheads agree; “Abs. error” compares reconstructed and uncut observables.

Circuit	Cuts	Pred. Γ	Exec. Γ	Abs. error
QAOA $n=6$, native RZZ	2	33.9706	33.9706	1.39×10^{-17}
VQE $n=6$, CX	1	9	9	5.55×10^{-17}

exercises the log-overhead reporting path; it does not verify the floating-point SCIP lower bounds.

Independently, a shot-free statevector reconstruction check (Table X) confirms that predicted and Qiskit-derived overheads agree at machine precision, validating the cost accounting used throughout.

- [1] J. Preskill, Quantum computing in the NISQ era and beyond, *Quantum* **2**, 79 (2018).
- [2] T. Peng, A. W. Harrow, M. Ozols, and X. Wu, Simulating large quantum circuits on a small quantum computer,

- Physical Review Letters **125**, 150504 (2020).
- [3] S. Bravyi, G. Smith, and J. A. Smolin, Trading classical and quantum computational resources, *Physical Review X* **6**, 021043 (2016).

- [4] W. Tang, T. Tomesh, M. Suchara, J. Larson, and M. Martonosi, CutQC: Using small quantum computers for large quantum circuit evaluations, in *Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems* (2021) pp. 473–486.
- [5] M. M. Deza and M. Laurent, *Geometry of Cuts and Metrics* (Springer, 1997).
- [6] E. L. Lawler and D. E. Wood, Branch-and-bound methods: A survey, *Operations Research* **14**, 699 (1966).
- [7] V. Kaibel and M. E. Pfetsch, Packing and partitioning orbitopes, *Mathematical Programming* **114**, 1 (2008).
- [8] S. Brandhofer, I. Polian, and K. Krsulich, Optimal partitioning of quantum circuits using gate cuts and wire cuts, *IEEE Transactions on Quantum Engineering* **5**, 1 (2024).
- [9] J. Nakamura, T. Satoh, and S. Sanji, Improved sampling bounds and scalable partitioning for quantum circuit cutting beyond bipartitions, *Physical Review A* **112**, 042422 (2025).
- [10] F. J. Fröhler, Y. Stade, C. Ufrecht, D. D. Scherer, and R. Wille, Scalable circuit cutting: A framework for combined gate and wire cuts using gate groups (2026), accepted at the 2026 IEEE International Conference on Quantum Computing and Engineering (QCE), arXiv:2608.05287 [quant-ph].
- [11] A. M. Brańczyk, A. Carrera Vazquez, D. J. Egger, B. Fuller, J. Gacon, J. R. Garrison, J. R. Glick, C. Johnson, S. Joshi, E. Pednault, C. D. Pemmaraju, P. Rivero, I. Shehzad, and S. Woerner, Qiskit addon: Circuit cutting (2024), version 0.10.0 archived as Zenodo record 15102772.
- [12] K. Mitarai and K. Fujii, Constructing a virtual two-qubit gate by sampling single-qubit operations, *New Journal of Physics* **23**, 023021 (2021).
- [13] P. Virtanen, R. Gommers, T. E. Oliphant, *et al.*, SciPy 1.0: Fundamental algorithms for scientific computing in python, *Nature Methods* **17**, 261 (2020).
- [14] P. Sanders and C. Schulz, Think locally, act globally: Highly balanced graph partitioning, in *Experimental Algorithms*, Lecture Notes in Computer Science, Vol. 7933 (Springer, 2013) pp. 164–175.
- [15] N. Quetschlich, L. Burgholzer, and R. Wille, MQT bench: Benchmarking software and design automation tools for quantum computing, *Quantum* **7**, 1062 (2023).
- [16] L. Burgholzer and R. Wille, QCEC: A JKQ tool for quantum circuit equivalence checking, *Software Impacts* **7**, 100051 (2021).
- [17] C. Hojny, M. Besançon, K. Bestuzheva, *et al.*, The SCIP optimization suite 10.0 (2025), arXiv:2511.18580 [math.OC].
- [18] K. Mitarai and K. Fujii, Overhead for simulating a non-local channel with local channels by quasiprobability sampling, *Quantum* **5**, 388 (2021).
- [19] C. Piveteau, D. Sutter, and S. Woerner, Quasiprobability decompositions with reduced sampling overhead, *npj Quantum Information* **8**, 12 (2022).
- [20] A. Lowe, M. Medvidovic, A. Hayes, L. J. O’Riordan, T. R. Bromley, J. M. Arrazola, and N. Killoran, Fast quantum circuit cutting with randomized measurements, *Quantum* **7**, 934 (2023).
- [21] C. Ufrecht, M. Periyasamy, S. Rietsch, D. D. Scherer, A. Plinge, and C. Mutschler, Cutting multi-control quantum gates with ZX calculus, *Quantum* **7**, 1147 (2023).
- [22] C. Ufrecht, L. S. Herzog, D. D. Scherer, M. Periyasamy, S. Rietsch, A. Plinge, and C. Mutschler, Optimal joint cutting of two-qubit rotation gates, *Physical Review A* **109**, 052440 (2024).
- [23] L. Schmitt, C. Piveteau, and D. Sutter, Cutting circuits with multiple two-qubit unitaries, *Quantum* **9**, 1634 (2025).
- [24] C. Piveteau and D. Sutter, Circuit knitting with classical communication, *IEEE Transactions on Information Theory* **70**, 2734 (2024).
- [25] C. Piveteau, L. Schmitt, and D. Sutter, Circuit cutting with classical side information, *Physical Review Research* **7**, 033063 (2025).
- [26] L. Brenner, C. Piveteau, and D. Sutter, Optimal wire cutting with classical communication, *IEEE Transactions on Information Theory* **71**, 7742 (2025).
- [27] X. Li, V. Kulkarni, D. T. Chen, Q. Guan, W. Jiang, N. Xie, S. Xu, and V. Chaudhary, Efficient circuit wire cutting based on commuting groups, in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2024) pp. 117–123, arXiv:2410.20313 [quant-ph].
- [28] D. T. Chen, E. H. Hansen, X. Li, A. Orenstein, V. Kulkarni, V. Chaudhary, Q. Guan, J. Liu, Y. Zhang, and S. Xu, Online detection of golden circuit cutting points, in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2023) pp. 26–31.
- [29] G. Gentinetta, F. Metz, and G. Carleo, Overhead-constrained circuit knitting for variational quantum dynamics, *Quantum* **8**, 1296 (2024).
- [30] A. Carrera Vazquez, C. Tornow, D. Riste, S. Woerner, M. Takita, and D. J. Egger, Combining quantum processors with real-time classical communication, *Nature* **636**, 75 (2024).
- [31] P.-H. Chen, D.-W. Chiou, B.-H. Chen, and J.-H. R. Jiang, Enhanced quantum circuit cutting framework for sampling overhead reduction (2024), arXiv:2412.17704 [quant-ph].
- [32] A. W. Harrow and A. Lowe, Optimal quantum circuit cuts with application to clustered hamiltonian simulation, *PRX Quantum* **6**, 010316 (2025).
- [33] M. Hart, Optimal two-qubit gate-cutting cost and measures of nonlocality (2026), arXiv:2609.30141 [quant-ph].
- [34] Y. Idan, E. Zahavi, E. Mentovich, E. Cohen, and S. Zaks, Quantum circuit cutting: Complexity and optimization (2026), arXiv:2604.23700 [quant-ph].
- [35] W. Tang and M. Martonosi, ScaleQC: A scalable framework for hybrid computation on quantum and classical processors (2022), arXiv:2207.00933 [quant-ph].
- [36] S. Kan, Z. Du, M. Palma, S. A. Stein, C. Liu, W. Wei, J. Chen, A. Li, and Y. Mao, Scalable circuit cutting and scheduling in a resource-constrained and distributed quantum system, in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2024) pp. 1077–1088.
- [37] S. Kan, Y. Li, H. Wang, S. Mouradian, and Y. Mao, Circuit folding: Scalable and graph-based circuit cutting via modular structure exploitation, in *2025 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)* (2025) pp. 1–9.
- [38] M. Eslaminia and S. Le Beux, Reducing maximum subcircuits depth in quantum circuit cutting, *IEEE Transactions on Quantum Engineering* **7**, 3103209 (2026).
- [39] K. Mesman, Y. Tang, M. Moller, B. Chen, and S. Feld,

- MOSAIQC: Mixed-topology-aware optimization for scalable approximate noise-informed quantum circuit cutting (2026), arXiv:2607.18888 [quant-ph].
- [40] H. Ebi, S. Nishio, and T. Satoh, Treewidth-aware gate cut selection for reducing transpilation overhead on superconducting quantum devices (2026), arXiv:2605.29723 [quant-ph].
 - [41] D. Pal, R. Majumdar, P. V. Seshadri, A. Ray, and Y. Simmhan, Noise-aware selection of circuit cutting strategies under hardware noise non-uniformity (2026), arXiv:2604.24422 [quant-ph].
 - [42] M. Periyasamy, C. Ufrecht, D. D. Scherer, and W. Mauerner, CutReg: A loss regularizer for enhancing the scalability of QML via adaptive circuit cutting, in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)* (2025) pp. 99–103, arXiv:2506.14858 [quant-ph].
 - [43] H. Lian, J. Xu, Y. Zhu, Z. Fan, Y. Liu, and Z. Shan, Fast reconstruction algorithm based on HMC sampling, *Scientific Reports* **13**, 17773 (2023).
 - [44] R. Tamura, T. Kashimata, Y. Hamakawa, K. Tatsumura, and H. Imai, Decomposition of multi-qubit gates for circuit cutting (2026), arXiv:2603.26278 [quant-ph].
 - [45] R. Mengoni, W. Nadalin, M. Rennela, J. Rotureau, T. Darras, J. Laurat, E. Diamanti, I. Lavdas, *et al.*, Efficient gate reordering for distributed quantum compiling in data centers, *EPJ Quantum Technology* **13**, 65 (2026).
 - [46] P. T. Hung and H. Bui, hungpt-uef/CertiCut: CertiCut v1.0.0 — official submission release (2026), prior manuscript/software release.
 - [47] B. W. Kernighan and S. Lin, An efficient heuristic procedure for partitioning graphs, *Bell System Technical Journal* **49**, 291 (1970).